BAV23C

Dual high-voltage switching diodes

1 April 2023 **Product data sheet**

1. General description

Dual high-voltage switching diodes, encapsulated in a small SOT23 (TO-236AB) Surface-Mounted Device (SMD) plastic package.

2. Features and benefits

- High switching speed: $t_{rr} \leq 50$ ns
- Low leakage current
- Repetitive peak reverse voltage: $V_{RRM} \leq 250$ V
- Low capacitance: $C_d \leq 2$ pF
- Small SMD plastic package

3. Applications

- High-speed switching at high voltage
- High-voltage general-purpose switching

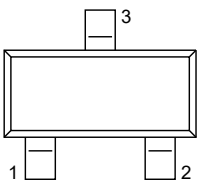
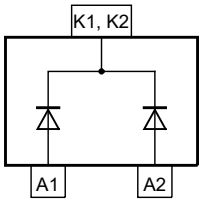
4. Quick reference data

Table 1. Quick reference data

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
Per diode							
I_R	reverse current	V_R = 200 V		-	-	100	nA
V_R	reverse voltage			-	-	200	V
t_{rr}	reverse recovery time	I_F = 10 mA; I_R = 10 mA; $I_{R(meas)}$ = 1 mA; R_L = 100 Ω; T_{amb} = 25 °C		-	-	50	ns

5. Pinning information

Table 2. Pinning information

Pin	Symbol	Description	Simplified outline	Graphic symbol
1	A1	anode (diode 1)	 SOT23	 aaa-027672
2	A2	anode (diode 2)		
3	K1, K2	common cathode		

6. Ordering information

Table 3. Ordering information

Type number	Package		
	Name	Description	Version
BAV23C	SOT23	plastic, surface-mounted package; 3 terminals; 1.9 mm pitch; 2.9 mm x 1.3 mm x 1 mm body	SOT23

7. Marking

Table 4. Marking codes

Type number	Marking code[1]
BAV23C	%V9

[1] % = placeholder for manufacturing site code

8. Limiting values

Table 5. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134).

Symbol	Parameter	Conditions		Min	Max	Unit
Per diode						
V_R	reverse voltage			-	200	V
V_{RRM}	repetitive peak reverse voltage			-	250	V
I_F	forward current	Single diode loaded	[1]	-	225	mA
			[2]	-	125	mA
I_{FRM}	repetitive peak forward current			-	625	mA
I_{FSM}	non-repetitive peak forward current	t_p = 1 µs; square wave	[3]	-	9	A
		t_p = 100 µs; square wave	[3]	-	3	A
		t_p = 10 ms; square wave	[3]	-	1.7	A
Per device						
P_{tot}	total power dissipation	$T_{amb} \leq$ 25 °C	[4]	-	250	mW
T_j	junction temperature			-	150	°C
T_{amb}	ambient temperature			-65	150	°C
T_{stg}	storage temperature			-65	150	°C

[1] Single diode loaded.
[2] Double diode loaded.
[3] T_j = 25 °C prior to surge.
[4] Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint.

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9. Thermal characteristics

Table 6. Thermal characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
Per device							
$R_{th(j-a)}$	thermal resistance from junction to ambient	in free air	[1]	-	-	500	K/W
$R_{th(j-sp)}$	thermal resistance from junction to solder point			-	-	360	K/W

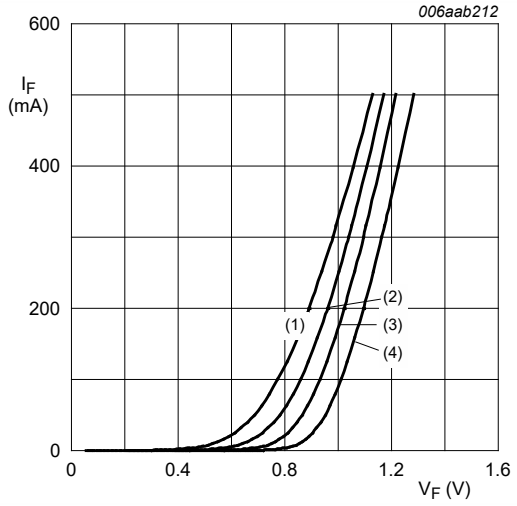
[1] Device mounted on an FR4 PCB, single-sided copper, tin-plated and standard footprint.

10. Characteristics

Table 7. Characteristics

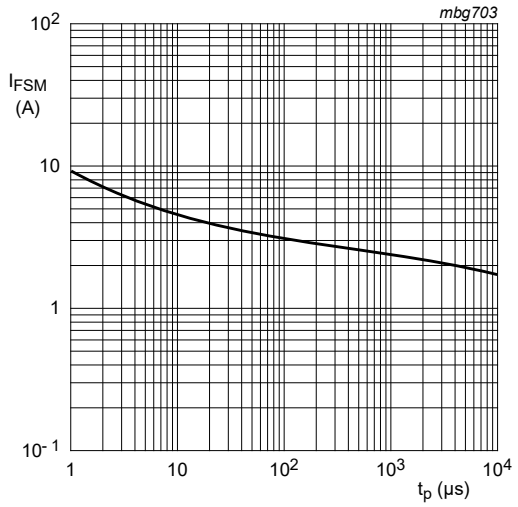
T_{amb} = 25 °C unless otherwise specified.

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
Per diode							
V_F	forward voltage	I_F = 100 mA		-	-	1	V
		I_F = 200 mA		-	-	1.25	V
I_R	reverse current	V_R = 200 V		-	-	100	nA
		V_R = 200 V; T_j = 150 °C		-	-	100	µA
C_d	diode capacitance	V_R = 0 V; f = 1 MHz		-	-	2	pF
t_{rr}	reverse recovery time	I_F = 10 mA; I_R = 10 mA; $I_{R(meas)}$ = 1 mA; R_L = 100 Ω; T_{amb} = 25 °C		-	-	50	ns



(1) T_{amb} = 150° C
(2) T_{amb} = 85° C
(3) T_{amb} = 25° C
(4) T_{amb} = -40° C

Fig. 1. Forward current as a function of forward voltage; typical values



Based on square wave currents.
$T_{j(init)}$ = 25 °C

Fig. 2. Non-repetitive peak forward current as a function of pulse duration; maximum values

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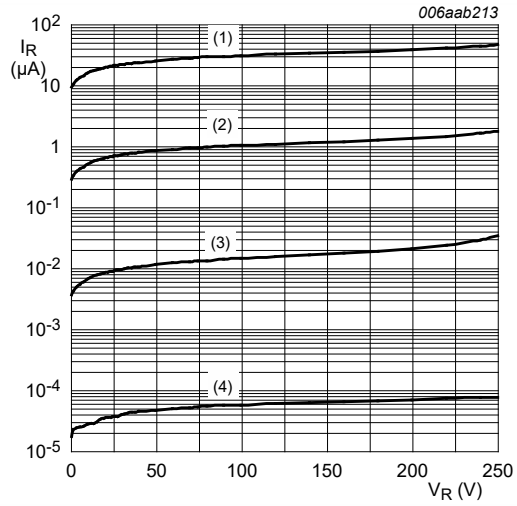
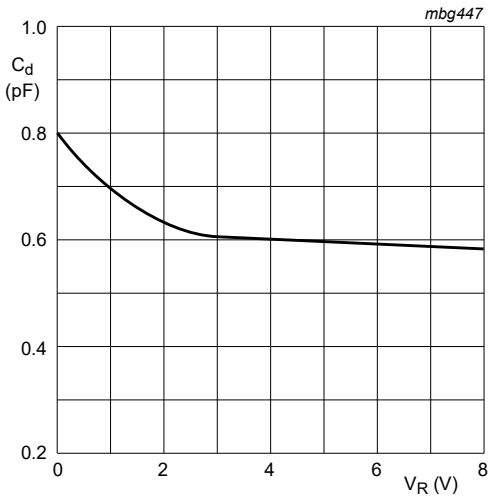


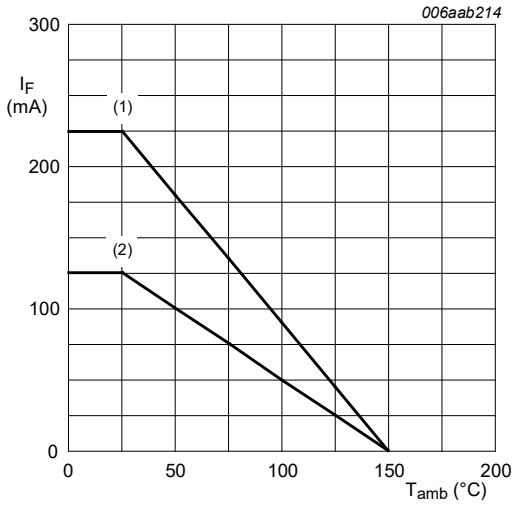
(1) T_{amb} = 150° C
(2) T_{amb} = 85° C
(3) T_{amb} = 25° C
(4) T_{amb} = -40° C

Fig. 3. Reverse current as a function of reverse voltage; typical values



f = 1 MHz
T_j = 25 °C.

Fig. 4. Diode capacitance as a function of reverse voltage; typical values.



FR4 PCB, standard footprint
(1) Single diode loaded
(2) Double diode loaded

Fig. 5. Forward current as a function of ambient temperature; derating curves

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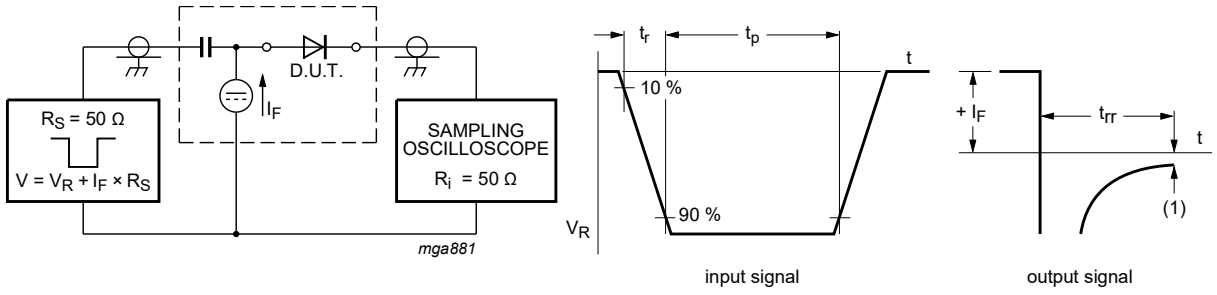
11. Test information



(1) I_R = 1 mA

Fig. 6. Reverse recovery time test circuit and waveforms

12. Package outline

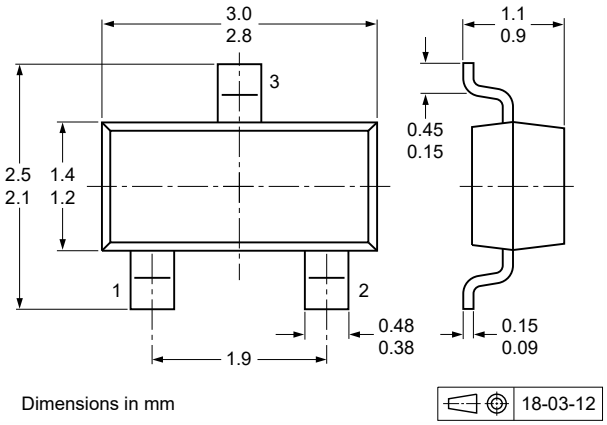


Fig. 7. Package outline SOT23

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13. Soldering

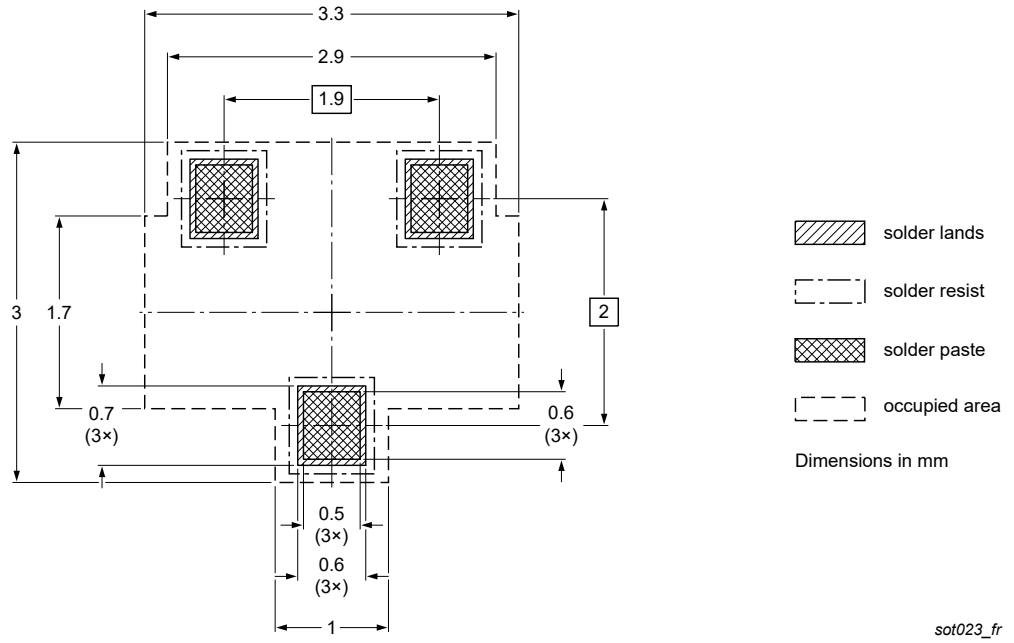

Fig. 8. Reflow soldering footprint for SOT23

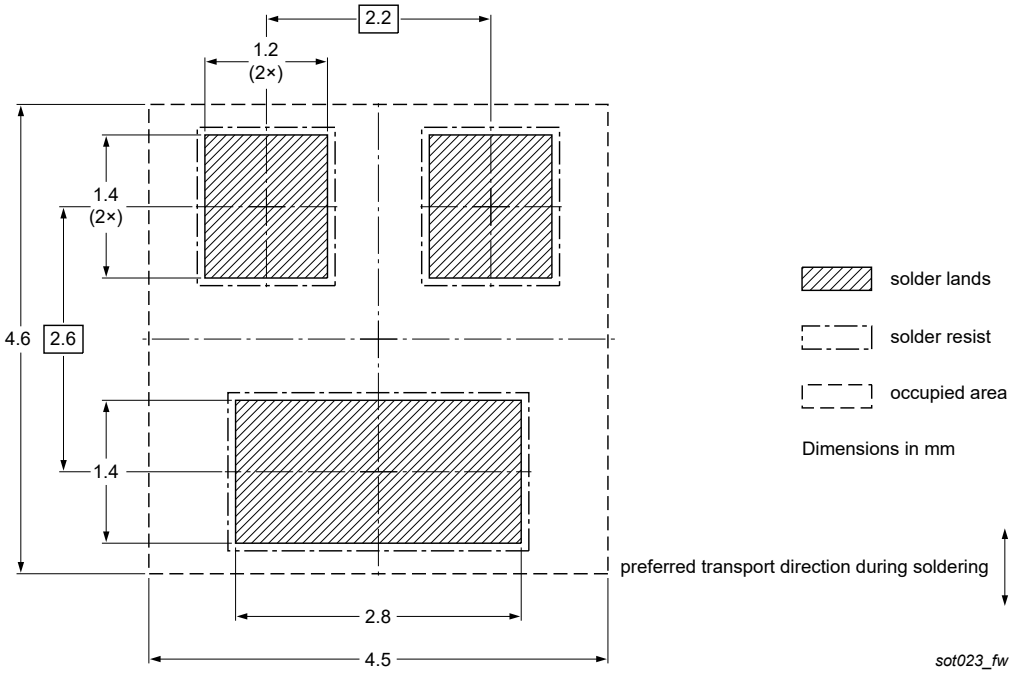

Fig. 9. Wave soldering footprint for SOT23

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14. Revision history

Table 8. Revision history

Data sheet ID	Release date	Data sheet status	Change notice	Supersedes
BAV23C v.8	20230401	Product data sheet	-	BAV23_SER_7
Modifications:	• Family data sheet reduced to single type data sheet. • The format of this data sheet has been redesigned to comply with the identity guidelines of Nexperia. • Legal texts have been adapted to the new company name where appropriate. • Product changed to non-automotive qualification. Please refer to nexperia.com for automotive (-Q) product alternative(s). • Packing information removed.			
BAV23_SER_7	20100319	Product data sheet	-	BAV23_SER_6
BAV23_SER_6	20080303	Product data sheet	-	BAV23S_5 BAV23_2
BAV23S_5	20011012	Product specification	-	BAV23S_4
BAV23_2	19960917	Product specification	-	BAV23_1

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15. Legal information

Data sheet status

Document status [1][2]	Product status [3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

[1] Please consult the most recently issued document before initiating or completing a design.
[2] The term 'short data sheet' is explained in section "Definitions".
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